

Meiru Zhang

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Core expertise: LLM Post-Training & Agentic RL (reward design, GRPO, PPO, RLVR), Long-Horizon Tool-Using Agents, Environment-Grounded Evaluation, Efficient Inference (NVFP4 / TensorRT), LLM Reliability & Interpretability

EDUCATION

- University of Cambridge**, PhD in Computation, Cognition and Language Oct 2021 – Aug 2026
- *Supervisor:* Prof. Nigel Collier | **Gates Cambridge Scholar** | *Research:* LLM Post-Training, Agentic RL, LLM Evaluation, RAG
 - *Thesis:* **Event-Centric AI: From Oracle-Free Event Extraction to Efficient Forecasting** — PhD completed and approved Aug 2026; conferral at a forthcoming Congregation.
 - *Research programme:* Independent work in Event-Centric AI — oracle-free event extraction, reliable use of retrieved evidence in long contexts, and efficient event forecasting; **four first-author papers** at ICLR, ACL and EMNLP.
- Imperial College London**, MSc in Computing (Machine Learning) Sept 2018 – Sept 2019
- **Distinction** | *Research:* Personalization, Active Learning, Reinforcement Learning
 - *Dissertation:* **Multimodal Active Learning from Human Data** — DQN-based active learning for data-scarce personalization.
- University College London**, BEng in Electronic and Electrical Engineering Sept 2015 – Jun 2018
- **Overall mark:** 81/100 (Top 5%) | **Dean's List** | Goldsmid Sessional Prize (2016, 2017)

WORK EXPERIENCE

- Epic Games, London, UK**, Machine Learning Intern (Part-time) Feb 2026 – Nov 2026
- **Long-horizon agentic RL for a coding agent:** Post-trained a 27B open-source model into a multi-step coding agent that calls VerseCoder APIs to resolve developer requests in a low-resource, domain-specific language (Verse). Built the training-data pipeline, the sandboxed trajectory harness, the reward and evaluation metrics its RL runs on, and an NVFP4 deployment via NVIDIA TensorRT Model Optimizer (ModelOpt) for efficient inference. **On internal test set, raised compile pass@1 from 50% to 80% and test-case correctness score from 0.51 to 0.84.**
 - **Environment-grounded agent evaluation:** Built executable evaluation for coding and visual agents. For the visual judge, annotated query-scene screenshot pairs, generated hard/easy adversarial negatives through cross-query pairing, and optimized the judge with GEPA; extended the end-to-end agent pipeline with adversarial tests that decompose queries and remove required features before scene generation, verifying that incomplete scenes receive low completeness scores. Now being integrated by Epic's engineering teams.
- Huawei Noah's Ark Lab, London, UK**, Research Intern in NLP Feb 2025 – Sept 2025
- **Problem:** Retrieval-augmented autoformalization fails when informal statements do not map cleanly to formal library concepts and retrieved premises give too little guidance on their use.
 - **Action:** Led method design and implementation of DRIFT: **Decompose** theorems into targeted sub-queries, **Retrieve** premises with a contrastively finetuned BGE-M3 dense retriever plus **Illustrative** usage examples, then **Formalize** Theorems.
 - **Result:** Up to 13.3 F1 gain on retrieval and 42.3% absolute gain on downstream autoformalization (BEq+@10) on the out-of-distribution ConNF benchmark. **First-author paper, ICLR 2026.**
- Antobot, UK**, Computer Vision Engineer Nov 2018 – Sept 2021
- Built the perception pipeline (detection, tracking, size estimation) for autonomous agricultural robots from scratch (no existing dataset): curated an instance-segmentation dataset and trained real-time models (MobileNetV2, DeepLabV3, YOLACT) plus yield/size regressors.
 - Production-deployed the on-device pipeline on mobile robotic platforms via TensorRT/ONNX for real-time edge inference ($\approx 57\%$ mAP@[.5:.95] on the custom dataset).

SELECTED PUBLICATIONS

- **Failure Modes in Multi-Hop QA: The Weakest Link Effect and the Recognition Bottleneck** **ACL 2026**
Meiru Zhang, Zaiqiao Meng, and Nigel Collier.
- **DRIFT: Decompose, Retrieve, Illustrate, then Formalize Theorems** **ICLR 2026**
Meiru Zhang, Philipp Borchert, Milan Gritta, and Gerasimos Lampouras.
- **The Power of Simplicity in LLM-Based Event Forecasting** **REALM Workshop, ACL 2025**
Meiru Zhang, Auss Abboud, Zaiqiao Meng, and Nigel Collier.
- **Can We Instruct LLMs to Compensate for Position Bias?** **EMNLP 2024**
Meiru Zhang, Zaiqiao Meng, and Nigel Collier.

- **BAND: Biomedical Alert News Dataset** AAI 2024
Zihao Fu, Meiru Zhang, Zaiqiao Meng, Yannan Shen, David Buckeridge, and Nigel Collier.
- **COFFEE: A Contrastive Oracle-Free Framework for Event Extraction** MATCHING Workshop, ACL 2023
Meiru Zhang, Yixuan Su, Zaiqiao Meng, Zihao Fu, and Nigel Collier.
- **Multi-modal Active Learning From Human Data: A Deep RL Approach** ICMI 2019
Ognjen Rudovic, Meiru Zhang, Bjorn Schuller, and Rosalind Picard.

SELECTED RESEARCH

LLM Evaluation: Inference-Time Steering & Reasoning Reliability (Attention Instruction to MFAI)

Dec 2023 – Jan 2026

- **Problem:** Position bias degrades long-context reasoning; prior mitigations need finetuning or architectural change, and multi-hop evaluations could not separate failure to *locate* evidence from failure to *reason* over it.
- **Action:** Developed **Attention Instruction** (EMNLP 2024), a training-free method to steer attention via natural language, then **Multi-Focus Attention Instruction (MFAI)** as a diagnostic probe; designed factorial evidence-placement experiments across 5 LLMs and two multi-hop QA tasks.
- **Result:** Discovered the *Weakest Link Effect* — multi-hop performance collapses to the least-visible evidence position (step-function, not linear decay); MFAI recovers up to 11.5% accuracy. Quantified a 6× cost–accuracy tradeoff for System-2 reasoning models (**ACL 2026**).

LLM-Based Event Forecasting with Structured RAG

Jul 2024 – Mar 2025

- **Problem:** Agentic (ReAct) forecasting is the default but is expensive, noisy and hard to explain.
- **Action:** Reframed the task as single-step retrieval-augmented generation with compact structured evidence (event distributions, temporal patterns); systematically compared agentic vs. single-step.
- **Result:** Structured RAG matched ReAct at ~10% of the token cost; unstructured context hurt smaller models. **REALM Workshop, ACL 2025**.

Oracle-Free Event Extraction (COFFEE)

May 2022 – Jan 2023

- **Problem:** Event-extraction systems depend on gold event types, triggers or hand-built templates that are unavailable in realistic deployment.
- **Action:** Introduced the Oracle-Free Event Extraction (OFEE) task and built a contrastive generator–selector framework that extracts structured events from document context alone.
- **Result:** Outperformed prior approaches under the oracle-free setting on two ACE05 variants and established a more realistic evaluation protocol. **MATCHING Workshop, ACL 2023**.

RL-Optimized Data Selection for Personalized Model Training

Mar 2019 – Aug 2019

- **Problem:** Personalizing multimodal models (face, body, audio, physiology) to new users is bottlenecked by annotation cost; each user’s distribution is unique and labels are expensive and subjective.
- **Action:** Trained a DQN policy to choose which points to label for finetuning per-user classifiers. State = classifier probability outputs (compact 16D vs. 378D raw); reward = +1 correct without label, −0.05 per label request, −1 incorrect without label. Group policy trained on existing users, then deployed to new users.
- **Result:** Outperformed random sampling, uncertainty sampling and feature-level fusion baselines. Published at **ICMI 2019** (collaboration with MIT Media Lab).

INDEPENDENT PROJECT

Priaxis — AI-Native Life-Management App with Forecaster

2025 – Present

- **Challenge:** Can an AI system forecast whether a user is progressing toward their goals, and surface meaningful reflection signals rather than just track tasks?
- **Approach:** Built an AI-native app that captures task trajectories (creation, modification, completion, reflection) as structured **event** data; the system forecasts task-completion likelihood, checks alignment with user values and generates personalized reflection signals. Built with the MCP protocol for structured tool use.

SKILLS

LLM / Post-Training: PyTorch, Transformers, verl, vLLM, TRL, LoRA & full finetuning, contrastive training, Weights & Biases

RL & Evaluation: DQN, PPO, RLVR, reward design, trajectory/sandbox harnesses, adversarial test generation, rubric-based eval, failure-mode analysis

Agent & RAG: LangChain, LangGraph, LangFuse, MCP protocol

Deployment: TensorRT, ONNX, AWS

Programming: Python, Java, C